

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 Total Marks 50 Duration :60 Minutes Annexure A
Scenario-Based Questions

Code of Conduct:

I M.Siddartha/1923111314 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Siddartha

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	

Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
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Scenario-Based Questions

1. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

- Interpret the scenario and explain the importance of boundary detection.
- Identify key challenges in detecting boundaries.
- Apply a suitable edge detection method.
- Justify your selection with logical reasoning.
- Present your explanation clearly and systematically.

ANSWER:

1. Object Boundary Detection

(a) Interpret the scenario and explain the importance of boundary detection

In a medical image, different tissues (such as organs, tumors, or bones) must be accurately separated. Boundary detection identifies the edges between adjacent tissues, helping doctors locate abnormalities, measure organ size, and support diagnosis and treatment planning.

(b) Key challenges in detecting boundaries

- Image noise causing false edges.
- Low contrast between neighboring tissues.
- Blurred or weak boundaries.
- Presence of overlapping structures.

(c) Suitable edge detection method

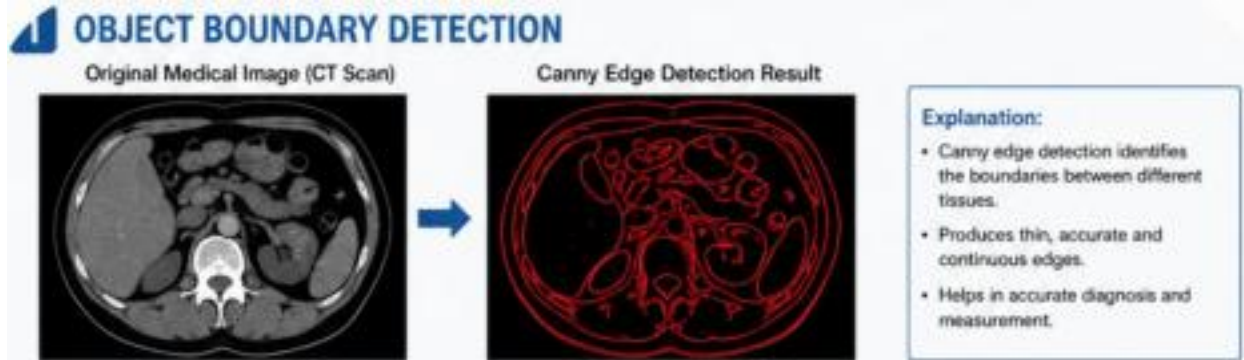
Canny Edge Detection

(d) Justification

- Effectively removes noise using Gaussian smoothing.
- Detects both strong and weak edges.
- Uses non-maximum suppression for thin edges.
- Employs hysteresis thresholding to produce continuous boundaries.

(e) Structured Explanation

- **Scenario:** Medical tissue boundary detection.
- **Problem:** Weak and noisy edges.
- **Method:** Canny Edge Detection.
- **Result:** Accurate, continuous, and reliable tissue boundaries.



2. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity. (a) Interpret the scenario and explain segmentation requirements.

(b) Identify key features enabling separation.

(c) Apply a suitable segmentation technique.

(d) Justify your decision logically.

(e) Present your response clearly.

ANSWER:

2. Simple Segmentation

(a) Interpret the scenario and explain segmentation requirements

The image contains objects that have intensity values clearly different from the background.

The goal is to separate the object from the background accurately.

(b) Key features enabling separation

- High intensity difference.
- Uniform background.
- Good image contrast.
- Minimal overlap between object and background intensities.

(c) Suitable segmentation technique

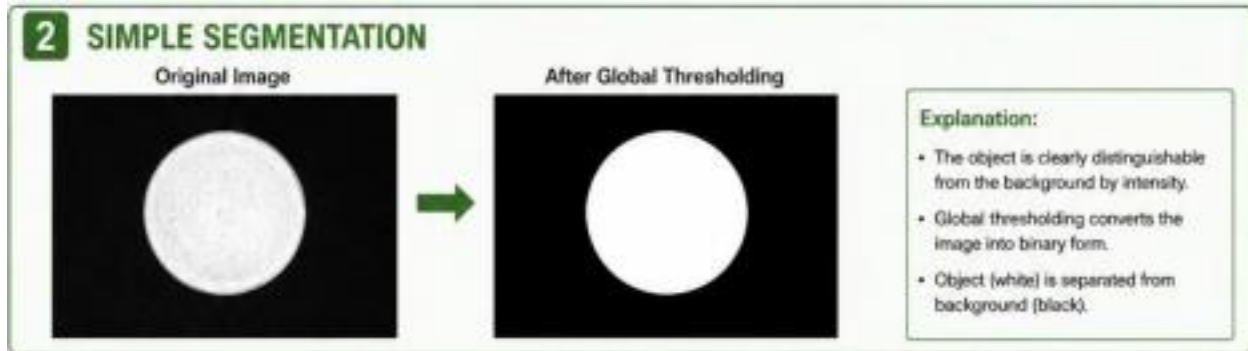
Global Thresholding (Binary Thresholding)

(d) Justification

- Simple and computationally efficient.
- Works well when object and background have distinct intensity values.
- Produces fast and accurate segmentation.

(e) Structured Explanation

- **Scenario:** Clear object-background separation.
- **Problem:** Segment object from background.
- **Method:** Global Thresholding.
- **Result:** Clean binary image with separated objects.



3. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

- Interpret the scenario and explain segmentation challenges.
- Identify key issues such as intensity overlap.
- Apply an appropriate advanced segmentation method.
- Justify how the method handles complexity.
- Provide a structured and clear explanation.

ANSWERS:

3. Complex Segmentation

(a) Interpret the scenario and explain segmentation challenges

The image contains multiple regions with similar or overlapping intensity values, making simple thresholding ineffective.

(b) Key issues

- Intensity overlap.
- Image noise.
- Irregular object shapes.
- Weak boundaries between neighboring regions.

(c) Appropriate advanced segmentation method

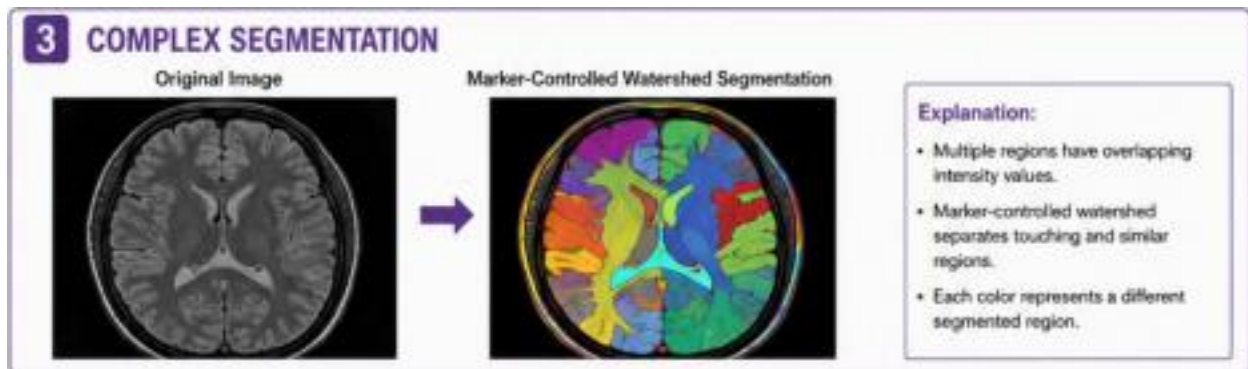
Watershed Segmentation (Marker-Controlled Watershed)

(d) Justification

- Separates touching and overlapping regions.
- Uses markers to reduce over-segmentation.
- Produces accurate region boundaries even with similar intensities.

(e) Structured Explanation

- **Scenario:** Multiple overlapping regions.
- **Problem:** Intensity overlap.
- **Method:** Marker-Controlled Watershed Segmentation.
- **Result:** Accurate separation of complex image regions.



4. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

- Interpret the scenario and explain why edges are broken.
- Identify key issues affecting boundary continuity.
- Apply a suitable method to improve edge connectivity.
- Justify your approach logically.
- Present your explanation clearly.

ANSWER:

4. Broken Edges

(a) Interpret the scenario and explain why edges are broken

After edge detection, some boundaries become discontinuous because of noise, weak gradients, or improper threshold settings.

(b) Key issues

- Noise.
- Weak edge responses.
- Incorrect threshold values.
- Missing edge pixels.

(c) Suitable method to improve connectivity

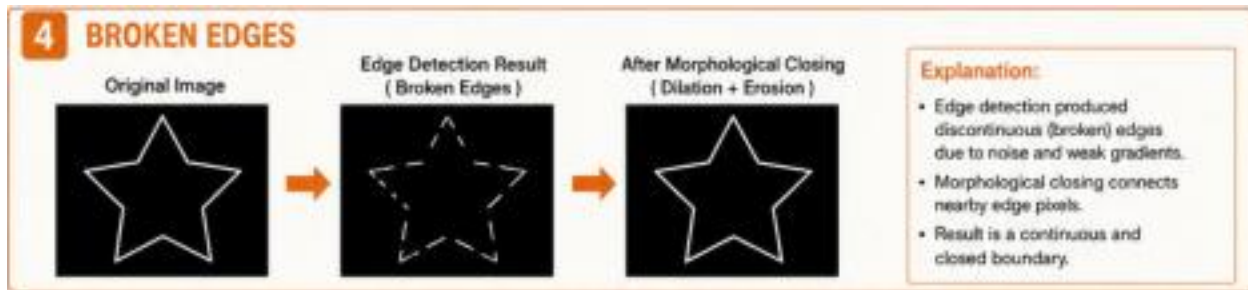
Morphological Closing (Dilation followed by Erosion)

(d) Justification

- Connects nearby edge segments.
- Fills small gaps in object boundaries.
- Preserves the overall object shape.
- Produces continuous edges for further processing.

(e) Structured Explanation

- **Scenario:** Discontinuous object boundaries.
- **Problem:** Broken edges.
- **Method:** Morphological Closing.
- **Result:** Continuous and connected object boundaries.



5. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

- Interpret the scenario and define the combined requirement.
- Identify key issues involving noise and edge preservation.
- Apply an appropriate sequence of techniques.
- Justify the order and selection with strong reasoning.
- Present your response in a clear and structured format.

ANSWER:

5. Combined Requirement

(a) Interpret the scenario and define the combined requirement

An industrial inspection system must remove image noise while preserving important edges so that defects such as cracks or scratches can be detected accurately.

(b) Key issues

- Noise may hide defects.
- Excessive smoothing may blur important edges.
- Edge preservation is essential for accurate inspection.

(c) Appropriate sequence of techniques

- Median Filter** (Noise Removal)
- Canny Edge Detection** (Edge Detection)

(d) Justification

- **Median Filter** removes impulse (salt-and-pepper) noise while preserving edges better than averaging filters.
- Applying **Canny Edge Detection** after noise removal produces accurate and continuous edges with fewer false detections.
- This sequence improves defect detection reliability.

(e) Structured Explanation

- **Scenario:** Industrial defect inspection.
- **Problem:** Remove noise without losing edges.
- **Technique Sequence:** Median Filter → Canny Edge Detection.
- **Result:** Noise-free image with sharp defect boundaries suitable for inspection.

5 COMBINED REQUIREMENT: NOISE REMOVAL WHILE PRESERVING EDGES

